FaultPilot: measured results and limits

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This brief transcribes the frozen repository results for paper preparation. It is not the required 8-10 page research paper and is not a Turnitin report.

Evaluation set

MLDebugging has 1,175 records. The source-task-grouped split contains 941 development records and 234 held-out test records. Original-code preflight found 198 reproducible failures, 30 setup errors, five timeouts, and one already passing case. Repair rates below use the fixed 198-case eligible denominator; all 234 records have traces.

Diagnosis

Method	Macro F1	Exact match
Phase 1 rules	0.04082	0.00427
Learned classifier	0.31054	0.08120
Full evidence + retrieval	0.31054	0.08120
Retrieval score fusion (alpha 0.2)	0.29062	0.15385

The learned classifier improved macro F1 by 0.26973 over Phase 1 rules (task-group bootstrap 95% interval 0.18266 to 0.35415). Retrieval examples were available for all 234 cases and local code-line evidence for 194, but availability does not establish fault localization accuracy.

Repair experiment

Mode	Passes / 198	Mean loop time	Key outcome
E2 Python base-paper adaptation	0 / 198	22.381 s	0 verified
E3 full FaultPilot	0 / 198	21.189 s	33 evidence abstentions
E5 no retrieval	1 / 198	22.466 s	One supplied-test pass
E5 no verification	0 / 198	17.951 s	123 proposals; 0 audit passes
E5 no evidence gate	0 / 198	10.170 s	0 verified

E2 is an adaptation of RepairAgent's inspect-edit-test loop to the same Python cases as E3. The original RepairAgent paper's reported 164 correct Java/Defects4J fixes are contextual and cannot be compared directly with these Python pass rates.

Ablation and failure accounting

Removing retrieval produced one supplied-test pass where E3 produced none. The paired pass-rate difference E3 minus no retrieval was -0.00505 (task-group bootstrap 95% interval -0.01596 to 0). Removing verification produced 123 proposals, none of which passed the separate offline test audit. The full system had 70 candidate load failures, 54 loaded candidates that failed tests, 40 rejected-edit cases, and 33 evidence abstentions. There were no successful E3 repairs to use as success examples.

Interpretation and limits

The diagnosis component improved category prediction; the repair loop did not fix a reproduced failure under the frozen budget and local 0.5B model. The evidence gate's conditional exact-match gain was only 0.00128, below the Phase 1 target of +0.10. A supplied-test pass does not prove semantic correctness. The benchmark supplies no independent regression suite; missing packages and timeouts restrict executable coverage. Mean loop times omit precomputed diagnosis and preflight and represent between-case, single-run measurements.

Repository evidence

Public repository: github.com/FazalZamanKhan/faultpilot-g2-08
Frozen release: github.com/FazalZamanKhan/faultpilot-g2-08/releases/tag/phase2-final